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Product Returns Analysis

1. Capture Usable Data

```
codio@biologylesson-ibizafloat:~/workspace$ mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 36
Server version: 5.5.62-0ubuntu0.14.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> show databases;
+----------------------+
| Database             |
+----------------------+
| information_schema   |
| QuantigrationRMA     |
| QuantigrationUpdates |
| classicmodels        |
| federle              |
| mysql                |
| performance_schema   |
+----------------------+
7 rows in set (0.00 sec)
```

```
mysql> use QuantigrationUpdates;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
```

```
mysql> show tables;
+--------------------------------+
| Tables_in_QuantigrationUpdates |
+--------------------------------+
| Collaborator                   |
| Customer                       |
| Orders                         |
| RMA                            |
+--------------------------------+
4 rows in set (0.00 sec)
```

In these screen shots I am just accessing the data, which I've already preloaded into Codio for further analysis. In the first screen shot I entered my sql. Next I wrote the command show databases so I could see the database I need to select. In the third screenshot I wrote the command use Quantigration Updates so I could use the data in that database. In the last screen shot I used the command show tables so I could see all the tables again and be familiar with the data.

2a. Analyze the number of returns by state

```
mysql> SELECT State, COUNT(RMA.RMA_ID) AS Product_Returned FROM Customer INNER JOIN Orders ON Customer.Cust
omer_ID = Orders.Order_ID INNER JOIN RMA ON Orders.Order_ID = RMA.Order_ID GROUP BY State ORDER BY Product_
Returned DESC;
```

State	Product_Returned
Massachusetts	385
West Virginia	338
Montana	336
Idaho	329
Tennessee	327
Delaware	323
Arkansas	322
Nevada	317
New Mexico	311
Oregon	311
Minnesota	309
Mississippi	309
New York	308
Arizona	305
Alabama	305
North Dakota	304
Washington	303
South Dakota	303
Maryland	303
New Hampshire	299
Iowa	298
Georgia	297
North Carolina	297
Michigan	296
California	296

Louisiana	296
Wyoming	295
Connecticut	295
Maine	294
South Carolina	293
Wisconsin	293
Rhode Island	293
Kentucky	292
Ohio	292
Indiana	292
Utah	291
Nebraska	290
Missouri	290
Florida	289
Pennsylvania	287
Texas	286
Colorado	286
Hawaii	284
Illinois	283
Vermont	277
New Jersey	268
Oklahoma	266
Kansas	249

48 rows in set (0.57 sec)

Query: SELECT State, COUNT (RMA.RMA_ID) AS Product_Returned FROM Customer INNER JOIN Orders ON Customer.Customer_ID = Orders.Order.ID INNER JOIN RMA ON Orders.Order_ID = RMA.Order_ID Group by State ORDER BY Product_Returned DESC;

In the screenshot above I used a select and inner join command to see the number of returns by state. From the resulting screenshot you can see that the top three returning states are Massachusetts, West Virginia and Montana. The states that return the least are New Jersey,

Oklahoma and Kansas. The number of returns for our entire operation range about 250 to 390 per state. There are no outstanding outliers in returns.

2b. Analyze the percentage of returns by product type

```
mysql> SELECT SKU AS Product_Type, COUNT(RMA.RMA_ID) AS Product_Returned, COUNT(RMA.RMA_ID)*100/(SELECT COU
NT(RMA_ID) FROM RMA) AS Percent_Returns FROM Orders INNER JOIN RMA ON Orders.Order_ID = RMA.Order_ID GROUP
BY SKU ORDER BY Product_Returned DESC;
```

```
+--------------+------------------+-----------------+
| Product_Type | Product_Returned | Percent_Returns |
+--------------+------------------+-----------------+
| BAS-48-1 C   |             8422 |         22.0691 |
| ENT-48-40F   |             6213 |         16.2806 |
| ENT-48-10F   |             4345 |         11.3857 |
| BAS-08-1 C   |             4308 |         11.2887 |
| ENT-24-10F   |             4295 |         11.2547 |
| ADV-48-10F   |             4197 |         10.9979 |
| ADV-24-10C   |             4189 |         10.9769 |
| ENT-24-40F   |             2159 |          5.6575 |
| BAS-24-1 C   |               34 |          0.0891 |
+--------------+------------------+-----------------+
9 rows in set (0.03 sec)
```

```
mysql> SELECT COUNT(RMA_ID) FROM RMA;
+---------------+
| COUNT(RMA_ID) |
+---------------+
|         38162 |
+---------------+
1 row in set (0.01 sec)
```

Query: SELECT SKU AS Product_Type, COUNT(RMA.RMA_ID) AS Product_Returned, COUNT(RMA.RMA_ID) * 100 / (SELECT COUNT (RMA_ID) FROM RMA) AS Percent_Returns FROM Orders INNER JOIN RMA ON Orders.Order_ID = RMA.Order_ID GROUP BY SKU ORDER BY Product_Returned DESC;

In the screenshots above I used a select and inner join command to see the percentage of returns by product types. In the second screen shot you can see that the top returned product is the BAS-48. It actually accounts for 22% of the total returns, which is pretty significant. The BAS-24 is the lowest returned product and only accounts for 0.09% of total products returned. In the third screen shot I just asked for the total number of returned products to make sure the math is correct for the percentages.

3. Report for stakeholders

I created two queries based on the data given to me to analyze merchandise returns for this company. The first query I created analyzed the number of returns by state. From the screenshot provided you can see that the top three returning states are Massachusetts, West Virginia and Montana. These three states each have returns over 330. The states that return the least are New Jersey, Oklahoma and Kansas. These three states are all below 270. The number of returns for our entire operation range about 250 to 390 per state. There are no outstanding outliers in returns, which indicates that no state is doing particularly well or worse off than any other. The product managers for the lower return states should be reviewed to see if any methods can be learned from these areas and applied elsewhere to improve profitability. The high return states should also be reviewed to see if there is anything that can be done to reduce the number of returns. I would suggest that the, number of total product sales are looked into for each state as well. This current data is a bit flawed since the number of sales could differ greatly affecting the average percentage of returns per state. I would look into these percentages to be able to get a more accurate read on the ability of the product managers per state. I would also suggest organizing the states into their regional departments so we can also see how the overall regions are doing.

The second query I used analyzed the percentage of returns by product type. The query is quite helpful since it lets you see which products maintain their likeability for consumers and have fewer manufacturing errors, which would constitute a return. In the screen shot provided for you, you can see that the top returned product is the BAS-48. It actually accounts for 22% of the total returns, which is pretty significant. I would suggest that the product manger look into this product to see if adjustments could be made to reduce the customer return rate. The BAS-24 is the lowest returned product and only accounts for 0.09% of total products returned. I would see if reports can be made to see why that is so we can improve the durability of our other products. I would keep in mind that the data presented here has the same flaw as earlier, in that, it does not take into account the number of product sales for each product type. I would suggest looking at the ratio of returns versus sales for each product type to have a better idea of the total amount of returns per product. I would also consider analyzing the number of returns on a state basis specifically for the BAS-48. This would let us collect more detailed data on this product to see if the percentage is average for each state or if it's being affected disproportionally in a specific region.